The Multiplicity of Meaning, Appendix A

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10th December 2024

1 Matters arising from earlier weeks

2 Fregeanism

Kit raised the question how Prior's paradox looks from a Fregean perspective, where words like 'assert' denote relations people bear to *thoughts*—and thoughts are not the *denotations* of sentences.[1]

[1] Note that one could agree with Frege about this without agreeing with him that there are only two possible denotations for sentences to have. Call the denotations of sentences—type t—'states of affairs'.

There are two possible ways for a Fregean to reconstruct a sentence like '$\exists p$. Mr X asserts p': they could treat this as a special case of quantifying in (wheeling in some apparatus that we know they'll need, to deal with '$\exists x$. Mr X asserts Fx', although they have been frustratingly silent about this), or they could treat the quantifier as quantifying over thoughts. Last week I assumed that they'd go for the former, but the latter is much more natural.

Here's an informal version. Let 't is a true thought' mean 'the state of affairs determined by t obtains' (we assume that every thought determines at most one state of affairs). We use the following schema:

Unique Determining The thought that P determines the state of affairs that P and no other state of affairs.

Given our definition of 'true thought', this implies

Truth Schema for Thoughts The thought that P is true iff P.

1. Mr X asserts the thought that not every thought asserted in Room 7 is true. (premise)
2. If every thought asserted in Room 7 is true, then the thought that not every thought asserted in Room 7 is true is true. (from 1)
3. Not every thought asserted in Room 7 is true. (From 2 and the Truth Schema.)
4. Some thought asserted in Room 7 is true. (From 1, 3 and the Truth Schema)

2.1 Prior's paradox with a universal quantifier

Suppose Mr X, in Room 7, produces an utterance that expresses that *every* proposition expressed by an utterance in Room 7 is false.

If every proposition expressed is false, then some proposition expressed is true. So, some proposition expressed by the utterance is true (in addition to the false proposition that every proposition expressed is false).

What are the true ones like? On a plausible compositional account of utterance-expressing[2] they are of the form *every proposition expressed* by an utterance in Room 7 is false*, where expressing* is a relation expressed by Mr X's token of 'express'.

[2] Note that such an account would not be so plausible for assertion—we assert certain consequences of propositions our utterances express.

For a proposition of this form to be true, there must be some cases where an utterance expresses a proposition it doesn't express*, since Mr X's utterance expresses some false propositions but doesn't express* any of them. It could be (though it need not be) that *expressing** is strictly stronger than *expressing*.

2.2 What are the other relations expressed by 'assert' and 'express' like?

The penultimate section of chapter 4 considers expressing for sentence *types*. It attempts to say something helpful in answer to the question 'What are relevant other relations expressed by "express" like?' The answer is that the relations expressed by 'express' differ in exactly the ways you'd think if you were just thinking about Sorites sequences—e.g. in where the draw the boundary between genuinely metaphorical uses (not expressed by the sentence) and mere idioms, or "frozen metaphors".

2.3 Could we have a disquotational truth predicate for sentences?

2.4 Consequences for LOT theories of the attitudes

3 The general theory of co-expressing

Higher-order background logic.

Type system: one non-terminal type e (individuals); three terminal types ε (object language expressions); κ (category-labels); t (propositions/states of affairs).

Syntactic constants:

$\mathrm{TBase} : \kappa \to t$	'…is a terminal base category'
$\mathrm{NBase} : \kappa \to t$	'…is a non-terminal base category'
$\mathrm{Const} : \kappa \to t$	'…is a constant with category …'
$\Rightarrow\, : \kappa \to \kappa \to \kappa$	'the type of functions from …to …'
$\circ : \varepsilon \to \varepsilon \to \varepsilon$	'the application of …to …'

Semantic constants:

$$\mathsf{E}_{\sigma_1,\ldots,\sigma_n} : \varepsilon^n \to \sigma_1 \to \cdots \to \sigma_n \to t$$
$$\mathsf{T}_\sigma : \kappa \to t$$

Semantic theory: syntax-independent axioms

Major inversion $\vec{a}, b\ \mathsf{E}_{\vec{\sigma},\tau}\ \vec{x}, y \to b, \vec{a}\ \mathsf{E}_{\tau,\vec{\sigma}}\ y, \vec{x}$

Minor inversion

$$\vec{a}, b, c\ \mathsf{E}_{\vec{\sigma},\tau,\rho}\ \vec{x}, y, z \to \vec{a}, c, b\ \mathsf{E}_{\vec{\sigma},\rho,\tau}\ \vec{x}, z, y$$

Deletion $\vec{a}, b\ \mathsf{E}_{\vec{\sigma},\tau}\ \vec{x}, y \to \vec{a}\ \mathsf{E}_{\vec{\sigma}}\ \vec{x}$

Duplication $\vec{a}, b\ \mathsf{E}_{\vec{\sigma},\tau}\ \vec{x}, y \to \vec{a}, b, b\ \mathsf{E}_{\vec{\sigma},\tau,\tau}\ \vec{x}, y, y$

Repetition $a, a\ \mathsf{E}_{\sigma,\sigma}\ x, y \to x =_\sigma y$

Co-interpretability $\mathsf{M}_\sigma b \wedge \vec{a}\ \mathsf{E}_{\vec{\tau}}\ \vec{x} \to \exists y^\sigma.\, b, \vec{a}\ \mathsf{E}_{\sigma,\vec{\tau}}\ y, \vec{x}$

where $\mathsf{M}_\sigma b := \exists x^\sigma . b\ \mathsf{E}_\sigma\ x$.

Syntax-dependent axioms:

Categories $\mathsf{T}_\sigma s \to \mathrm{Category}(s)$

Arrow $\mathsf{T}_{\sigma\to\tau}(s \Rightarrow t) \leftrightarrow \mathsf{T}_\sigma s \wedge \mathsf{T}_\tau t$

Type-constraint $\mathsf{M}_\sigma a \to \exists s^\kappa.\, a :: s \wedge \mathsf{T}_\sigma s$

Here, $\mathrm{Category}(s)$ (‘s^κ is a well-formed category’) and $a :: s$ (‘a is a well-formed expression of category s’) are defined in terms of the syntactic primitives.

Meaningfulness

$$\mathsf{T}_\sigma s \wedge \mathsf{T}_\tau t \wedge a :: s \Rightarrow t \wedge b :: s \to (\mathsf{M}_\tau a \circ b \leftrightarrow (\mathsf{M}_{\sigma\to\tau} a \wedge \mathsf{M}_\sigma b))$$

Application $a, b, a \circ b\ \mathsf{E}_{\sigma\to\tau,\sigma,\tau}\ x, y, z \to z =_\tau xy$

An illustrative theorem:

$$(b :: s \Rightarrow t \wedge c :: s) \to ((\vec{a}, b, c, b \circ c\ \mathsf{E}_{\vec{\rho},\sigma\to\tau,\sigma,\tau}\ \vec{x}, y, z, u) \leftrightarrow (\vec{a}, b, c\ \mathsf{E}_{\vec{\rho},\sigma\to\tau,\sigma}\ \vec{x}, y, z \wedge u =_\tau yz))$$

Using this (and Repetition) we can “reduce” any claim about what some list of given complex expressions co-express to a claim about what a nonrepeating list of constituent atomic expressions co-express.

Also, if we define a ‘semantic uniqueness’ predicate U_σ in the obvious way[3], we can show that semantically unique elements can always be pulled out of the list:

$$U_\tau b \to (\vec{a}, b\ \mathsf{E}_{\vec{\sigma},\tau}\ \vec{x}, y \leftrightarrow (\vec{a}\ \mathsf{E}_{\vec{\sigma}}\ \vec{x} \wedge b\ \mathsf{E}_\tau\ y))$$

[3] $aU_\sigma x := \exists x^\sigma . \forall y^\sigma .\, a\ \mathsf{E}_\sigma\ x \leftrightarrow x =_\sigma y$

4 *Extension to variables*

With Functionality: Suppose our background logic/metaphysics includes the following principle:

Functionality $\forall X^{\sigma\to\tau} Y^{\sigma\to\tau} . (\forall z^\sigma . Xz =_\tau Yz) \to X =_{\sigma\to\tau} Y$

Then there’s a neat way to extend this system to include expressions with free variables and variable-binders, based on the idea that a variable in a certain category expresses everything of the corresponding type, independent of any other expressions in which that variable doesn't occur free.

New syntactic constants: $\mathsf{V}_\sigma : \varepsilon \to t$ ('is a variable'), $\underline{\lambda} : \varepsilon \to \varepsilon \to \varepsilon$ ('the result of combining with a lambda').

Variables
$$\mathsf{V}_\rho b \wedge \bigwedge_i \neg\,\mathrm{FreeIn}(b, a_i) \wedge \vec{a}\ \mathsf{E}_{\vec{\sigma}}\ \vec{x} \to b, \vec{a}\ \mathsf{E}_{\rho,\vec{\sigma}}\ y, \vec{x}$$

Abstraction
$$\mathsf{V}_\sigma b \to \bigwedge_i \neg\,\mathrm{FreeIn}(b, a_i) \to (\forall x^\sigma y^\tau .\, b, c, \vec{a}\ \mathsf{E}_{\sigma,\tau,\vec{\sigma}}\ x, y, \vec{z} \leftrightarrow y = wx) \to (\forall z^{\sigma\to\tau} .\, \underline{\lambda} bc, \vec{a}\ \mathsf{E}_{\sigma\to\tau,\vec{\sigma}}\ z, \vec{z} \leftrightarrow z = w)$$

Without Functionality: If we couldn't think of any other decent way to handle variables, there might be an interesting argument for Functionality. But in fact, there is an alternative approach, on which the fundamental meaning-bearers are not just *expressions* but 'typings' or *expressions-in-a-context.*[4]

[4] 'Context' here means 'finite orderd non-repeating list of variables'. For a typing of an expression to be well formed, all the free variables in the expression must appear somewhere in the context.

(In the chapter I sketch this out in a way that probably unnecessarily complex, since it's simultaneously switching to a syntax where variables don't have a 'built in' type but rather have to have their types "declared" as part of the context.)

5 *Co-expressing for expression-tokens*

One thing I want to think more about is how the ideology of co-expressing should work when we are talking about expression-*tokens*.

One important feature is that speakers' intentions can generate co-ordination constraints even between tokens of semantically independent types.

(1) This is a cup. (Used to say only truths)

(2) That fact (*gesturing vaguely*) could not have obtained without him having eaten a sandwich.

I'm undecided about whether I have any use for a contrast between co-ordinate and indeendent expressing for word-tokens belonging to different sentence-tokens—even produced by different speakers.[5]

[5] I'm pretty sure I *don't* want to talk about co-ordinate expressing 'across possible worlds'.

6 *Generalizations for natural languages*

1. Non-uniform interpretation.
2. Richer syntax.

— Meaning-bearers as functions?

3. Type ambiguity.